



MHH772 / 90DaysOfDevOps



Code Pull requests Agents Actions Projects Wiki Security and quality Insights

Settings



master

90DaysOfDevOps / 2026 / day-07



LondheShubham153 added day 7 task

0be4b77 · 7 months ago



This branch is 6 commits ahead of TrainWithShubham/90DaysOfDevOps:master



Contribute



Sync fork

Name

Name

Last commit date



..



README.md

added day 7 task

7 months ago

README.md



# Day 07 – Linux File System Hierarchy & Scenario-Based Practice

## Task

Today's goal is to **understand where things live in Linux** and **practice troubleshooting like a DevOps engineer**.

You will create notes covering:

- Linux File System Hierarchy (the most important directories)
- Practice solving real-world scenarios step by step

This consolidates your Linux fundamentals and prepares you for real-world troubleshooting.

# Expected Output

---

By the end of today, you should have:

- A markdown file named: `day-07-linux-fs-and-scenarios.md`

or

- A hand written set of notes (Recommended)

Your notes should have two sections: File System Hierarchy and Scenario Practice.

## Guidelines

---

### Part 1: Linux File System Hierarchy (30 minutes)

Document the purpose of these **essential** directories:

#### Core Directories (Must Know):

- `/` (root) - The starting point of everything
- `/home` - User home directories
- `/root` - Root user's home directory
- `/etc` - Configuration files
- `/var/log` - Log files (very important for DevOps!)
- `/tmp` - Temporary files

#### Additional Directories (Good to Know):

- `/bin` - Essential command binaries
- `/usr/bin` - User command binaries
- `/opt` - Optional/third-party applications

For each directory:

- Write 1-2 lines explaining what it contains
- Run `ls -l <directory>` and note 1-2 files/folders you see
- Write one sentence: "I would use this when..."

#### Hands-on task:

```
# Find the largest log file in /var/log
du -sh /var/log/* 2>/dev/null | sort -h | tail -5
```



```
# Look at a config file in /etc
cat /etc/hostname

# Check your home directory
ls -la ~
```

## Part 2: Scenario-Based Practice (40 minutes)

**Important:** Focus on understanding the **troubleshooting flow**, not memorizing commands. Use the hints!

### SOLVED EXAMPLE: Understanding How to Approach Scenarios

#### Example Scenario: Check if a service is running

Question: How do you check if the 'nginx' service is running?



#### My Solution (Step by step):

##### Step 1: Check service status

```
systemctl status nginx
```



**Why this command?** It shows if the service is active, failed, or stopped

##### Step 2: If service is not found, list all services

```
systemctl list-units --type=service
```



**Why this command?** To see what services exist on the system

##### Step 3: Check if service is enabled on boot

```
systemctl is-enabled nginx
```



**Why this command?** To know if it will start automatically after reboot

**What I learned:** Always check status first, then investigate based on what you see.

Now try these scenarios yourself:

## Scenario 1: Service Not Starting

A web application service called 'myapp' failed to start after a server reboot.



What commands would you run to diagnose the issue?

Write at least 4 commands in order.

### Hint:

- First check: Is the service running or failed?
- Then check: What do the logs say?
- Finally check: Is it enabled to start on boot?

**Commands to explore:** `systemctl status myapp`, `systemctl is-enabled myapp`, `journalctl -u myapp -n 50`

**Resource:** Review Day 04 (Process and Services practice)

### Template for your answer:

Step 1: [command]

Why: [one line explanation]



Step 2: [command]

Why: [one line explanation]

...

## Scenario 2: High CPU Usage

Your manager reports that the application server is slow.



You SSH into the server. What commands would you run to identify which process is using high CPU?

### Hint:

- Use a command that shows **live** CPU usage
- Look for processes sorted by CPU percentage
- Note the PID (Process ID) of the top process

**Commands to explore:** `top` (press 'q' to quit), `htop`, `ps aux --sort=-%cpu | head -10`

**Resource:** Review Day 05 (Troubleshooting Drill - CPU & Memory section)

## Scenario 3: Finding Service Logs

A developer asks: "Where are the logs for the 'docker' service?"  
The service is managed by systemd.  
What commands would you use?



### Hint:

- systemd services → logs are in journald
- Command pattern: `journalctl -u <service-name>`
- Use `-n` flag to limit number of lines
- Use `-f` flag to follow logs in real-time (like `tail -f`)

### Commands to explore:

```
# Check service status first
systemctl status ssh
```

```
# View last 50 lines of logs
journalctl -u ssh -n 50
```

```
# Follow logs in real-time
journalctl -u ssh -f
```



**Resource:** Review Day 04 (Process and Services - Log checks section)

## Scenario 4: File Permissions Issue

A script at `/home/user/backup.sh` is not executing.  
When you run it: `./backup.sh`  
You get: "Permission denied"



What commands would you use to fix this?

### Hint:

- First: Check what permissions the file has
- Understand: Files need 'x' (execute) permission to run
- Fix: Add execute permission with `chmod`

### Step-by-step solution structure:



Step 1: Check current permissions

Command: `ls -l /home/user/backup.sh`

Look for: `-rw-r--r--` (notice no 'x' = not executable)

Step 2: Add execute permission

Command: `chmod +x /home/user/backup.sh`

Step 3: Verify it worked

Command: `ls -l /home/user/backup.sh`

Look for: `-rwxr-xr-x` (notice 'x' = executable)

Step 4: Try running it

Command: `./backup.sh`

**Resource:** Review Day 02 (File Permissions and Users Management)

## Why This Matters for DevOps

---

Understanding the file system is critical for:

- Knowing where to find logs, configs, and binaries
- Troubleshooting deployment issues
- Writing automation scripts that work across systems

Scenario-based practice prepares you for:

- Real production incidents
- DevOps interviews
- On-call troubleshooting under pressure

These are questions you **will** face in interviews and during real incidents.

## Submission

---

1. Fork this `90DaysOfDevOps` repository
2. Navigate to the `2026/day-07/` folder
3. Add your `day-07-linux-fs-and-scenarios.md` file
4. Commit and push your changes to your fork

## Learn in Public

---

Share your Day 07 progress on LinkedIn:

- Post 2–3 lines on what you learned about Linux file system
- Share one scenario you found challenging and how you solved it
- Optional: screenshot of your notes

Use hashtags:

#90DaysOfDevOps

#DevOpsKaJosh

#TrainWithShubham



Happy Learning **TrainWithShubham**



 master ▾

[90DaysOfDevOps](#) / [2026](#) / [day-07](#) /



↑ Top